State Estimation and Filtering for Robotic Perception

AA273

Extended Kalman Filter, Unscented Kalman Filter



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 4 due Fri Feb 13
- Group signup sheet posted (groups from 1 – 4 people)
- Midterm: Feb 5-7
  - 5 hour timed take home exam, you choose any 5 hours in 48 hour window **starting 3pm today**
  - Open book, notes, lecture videos
  - No internet, gen AI, or collaboration, coding assistants are allowed.
- Project Proposal: due Tue Feb 17
  - 2-3 pages, Latex
  - Related work >= 10 refs, ~1 page
  - Proposed work ~ 1 page
  - Bibtex refs

Lecture 10:

- Extended Kalman Filter (EKF)
- Derivation
- Unscented transform
- Unscented Kalman filter

Recall: L-G SS Model and KF Eqns

Dynamics eqn.:

$$X_{t+1} = A_t X_t + B_t u_t + W_t$$

Measurement eqn.:

$$Y_t = C_t X_t + D_t u_t + V_t$$

Initial condition:

$$X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$$

Whiteness and Indep.:

$$W_t \sim GWN(0, Q_t)$$

$$V_t \sim GWN(0, R_t)$$

X_0, W_t, V_τ independent for all t, τ

Given: $\mu_{0|0}, \Sigma_{0|0},\ A_t, B_t, C_t,\ Q_t, R_t$

(Assume $D_t = 0$)

For: $t = 1, 2, \ldots$

Prediction:

$$\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

Update:

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$$

$$K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$$

Loop

Nonlinear SS Model and Extended Kalman Filter

Dynamics eqn.:

$$X_{t+1} = f(X_t, u_t) + W_t$$

Measurement eqn.:

$$Y_t = g(X_t) + V_t$$

Uncorrelated IC, and noise processes:

$X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$

$W_t \sim GWN(0, Q_t)$

$V_t \sim GWN(0, R_t)$

X_0, W_t, V_τ independent for all t, τ

Jacobian matrices A_t and C_t

$$A_t = \nabla_{x_t} f(\mu_{t|t}, u_t)$$

$$C_t = \nabla_{x_t} g(\mu_{t|t-1})$$

Given: $\mu_{0|0}, \Sigma_{0|0}, f(\cdot,\cdot), g(\cdot), Q_t, R_t$

For: $t = 1, 2, \ldots$

Prediction:

$$\mu_{t|t-1} = f(\mu_{t-1|t-1}, u_{t-1})$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

Update:

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - g(\mu_{t|t-1}))$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$$

$$K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$$

Loop

Brief history

- 1960 – Rudolf Kalman. Trans. ASME, Journal of Basic Eng. Kalman Filter.
- 1961 – R. Kalman, R. S. Bucy, Trans. ASME JBE. Kalman-Bucy filter (cont. time),
- 1962 - G.L. Smith; S.F. Schmidt and L.A. McGee. NASA Ames tech report.
- 1993 – G. Kitagawa *Proc. of U.S.-Japan Joint Seminar on Statistical Time Series Analysis*. "Monte Carlo Filter"
- 1997 – H. Carvalho,P. Del Moral, A. Monin, G. Salut, Gérard. *IEEE Transactions on Aerospace and Electronic Systems*
- 1997 – S. J. Julier, J. K. Uhlmann. In *Proc. of Intl Symp on Aerospace/Defense Sensing, Simulation and Controls*.
- 2010s – Optimization based approaches, graphical models, factor graphs

Jacobians computation

$$f(x_t, u_t) = \begin{bmatrix} f_1(x_t, u_t) \\ \vdots \\ f_n(x_t, u_t) \end{bmatrix} \qquad x_t = \begin{bmatrix} x_t^1 \\ \vdots \\ x_t^n \end{bmatrix}$$

Partial of i th element of f()

With respect to j th element of x

$$A_t = \nabla_{x_t} f(\mu_{t|t}, u_t) = \begin{bmatrix} \frac{\partial f_1(x_t,u_t)}{\partial x_t^1} \Big|_{x_t=\mu_{t|t}} & \frac{\partial f_1(x_t,u_t)}{\partial x_t^2} \Big|_{x_t=\mu_{t|t}} & \cdots & \frac{\partial f_1(x_t,u_t)}{\partial x_t^n} \Big|_{x_t=\mu_{t|t}} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial f_n(x_t,u_t)}{\partial x_t^1} \Big|_{x_t=\mu_{t|t}} & \frac{\partial f_n(x_t,u_t)}{\partial x_t^2} \Big|_{x_t=\mu_{t|t}} & \cdots & \frac{\partial f_n(x_t,u_t)}{\partial x_t^n} \Big|_{x_t=\mu_{t|t}} \end{bmatrix}$$

$$A_t \in \mathbb{R}^{n \times n}$$

$$g(x_t) = \begin{bmatrix} g_1(x_t) \\ \vdots \\ g_m(x_t) \end{bmatrix}$$

Partial of i th element of g()

With respect to j th element of x

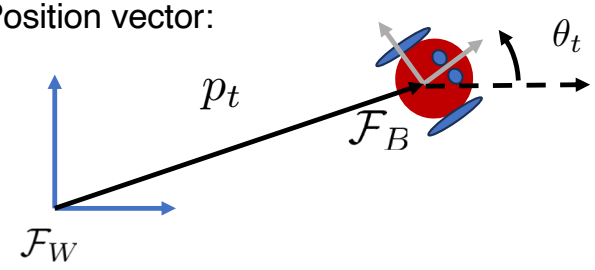
$$C_t = \nabla_{x_t} g(\mu_{t|t-1}) = \begin{bmatrix} \frac{\partial g_1(x_t)}{\partial x_t^1} \Big|_{x_t=\mu_{t|t-1}} & \frac{\partial g_1(x_t)}{\partial x_t^2} \Big|_{x_t=\mu_{t|t-1}} & \cdots & \frac{\partial g_1(x_t)}{\partial x_t^n} \Big|_{x_t=\mu_{t|t-1}} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial g_m(x_t)}{\partial x_t^1} \Big|_{x_t=\mu_{t|t-1}} & \frac{\partial g_m(x_t)}{\partial x_t^2} \Big|_{x_t=\mu_{t|t-1}} & \cdots & \frac{\partial g_m(x_t)}{\partial x_t^n} \Big|_{x_t=\mu_{t|t-1}} \end{bmatrix}$$

$$C_t \in \mathbb{R}^{m \times n}$$

Examples: Differential Drive Robot

Heading angle:

Position vector:



Robot state:

$$x_t = \begin{bmatrix} p_t^x \\ p_t^y \\ \theta_t \end{bmatrix}$$

Control input:

$$u_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}$$

Robot dynamics model cont. time:

$$\begin{aligned} \dot{p}_t^x &= v_t \cos\theta_t \\ \dot{p}_t^y &= v_t \sin\theta_t \\ \dot{\theta}_t &= \omega_t \end{aligned}$$

Robot dynamics model discrete. time with noise:

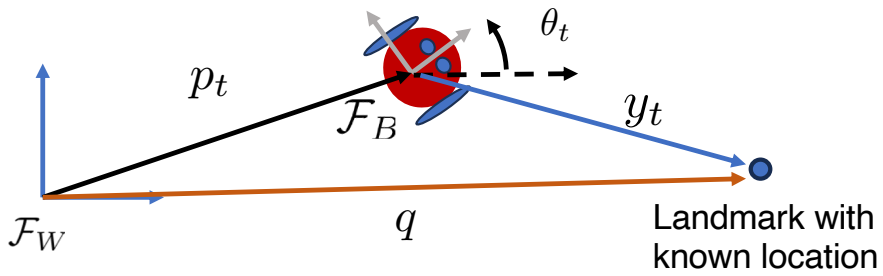
$$\begin{aligned} p_{t+1}^x &= p_t^x + \delta_t v_t \cos\theta_t + w_t^x \\ p_{t+1}^y &= p_t^y + \delta_t v_t \sin\theta_t + w_t^y \\ \theta_{t+1} &= \theta_t + \delta_t \omega_t + w_t^\theta \end{aligned}$$

Compute dynamics Jacobian:

$$A_t = \nabla_{x_t} f(\mu_{t|t}, u_t)$$

On the board, in the notes

Examples: Range Sensor Model



Range sensing model:

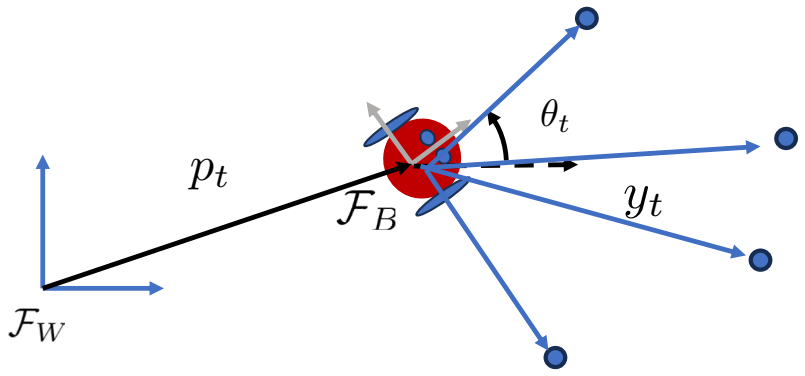
$$y_t = \|p_t - q\| + v_t$$

Compute measurement Jacobian:

$$C_t = \nabla_{x_t} g(\mu_{t|t-1})$$

On the board, in the notes

Examples: Range Sensor Model



Range sensing model for each landmark i:

$$y_t^i = \|p_t - q_i\| + v_t^i$$

Compute measurement Jacobian for each landmark:

$$C_t^i = \nabla_{x_t} g^i(\mu_{t|t-1})$$

With multiple landmarks, just stack measurements:

$$y_t = \begin{bmatrix} y_t^1 \\ \vdots \\ y_t^N \end{bmatrix}$$

Full Jacobian is just stacked Jacobians for each landmark:

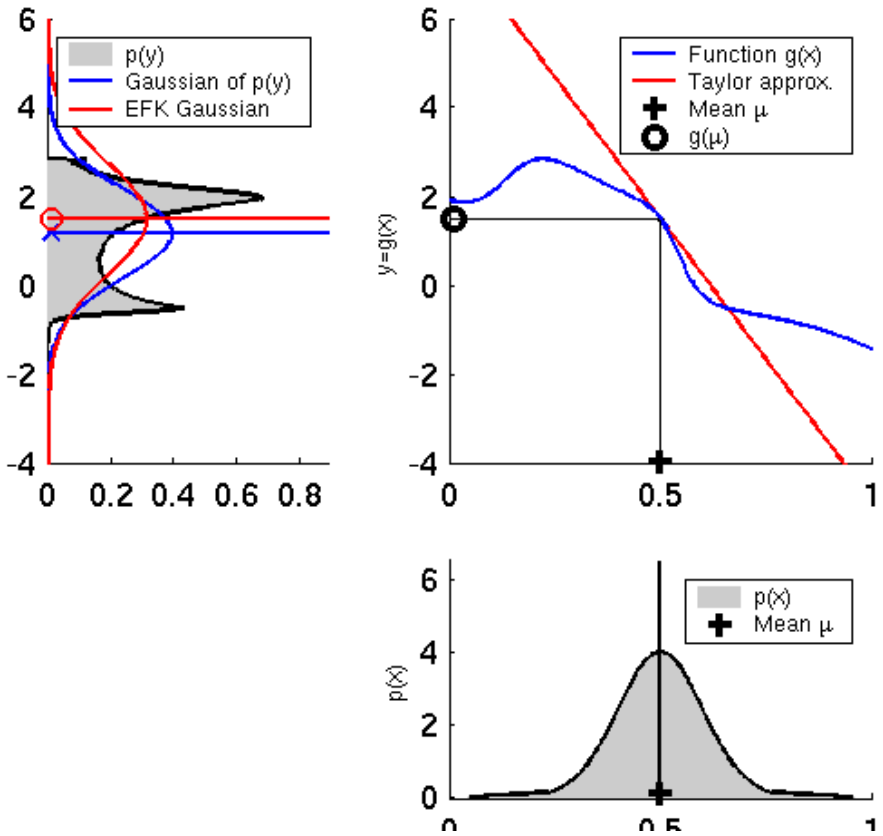
$$C_t = \begin{bmatrix} C_t^1 \\ \vdots \\ C_t^N \end{bmatrix}$$

This is called EKF localization.

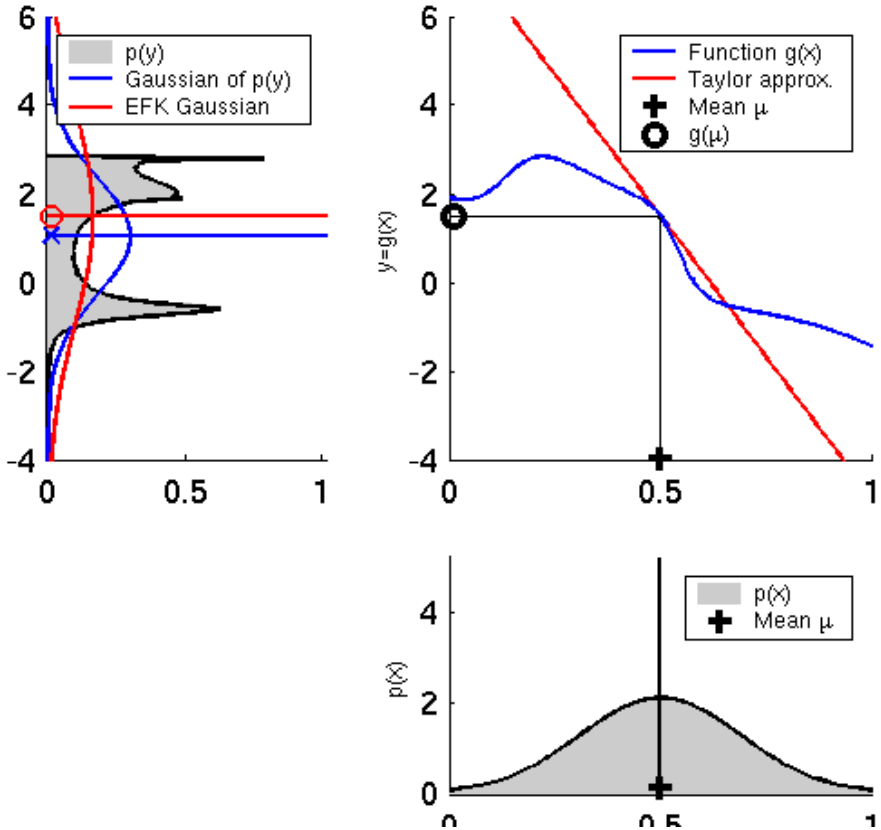
Towards a Derivation of the EKF

On the board, in the notes.

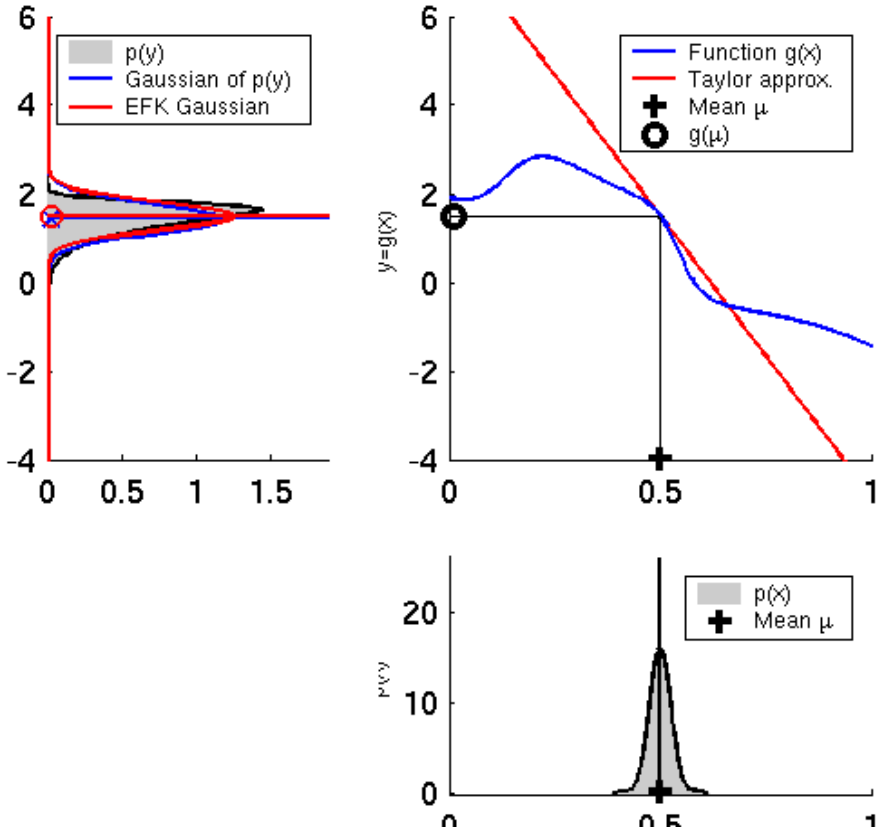
EKF: intuition



EKF: intuition

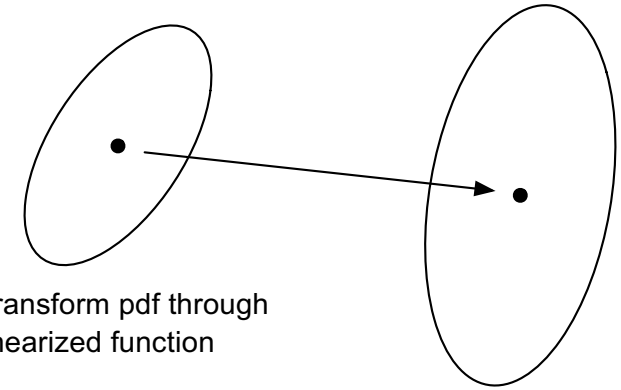


EKF: intuition



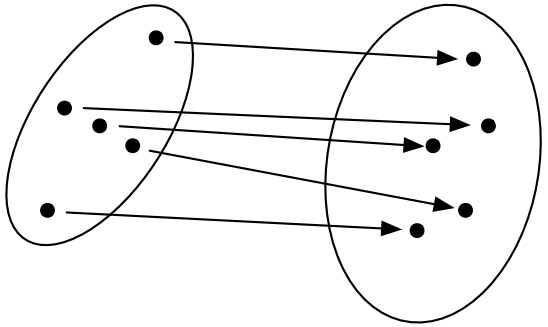
Unscented Kalman filter (UKF) – basic idea

EKF



Transform pdf through
linearized function

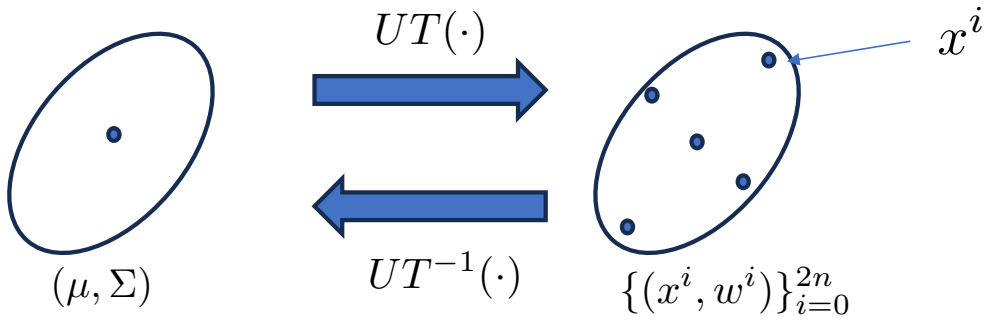
UKF



1. Compute sigma-points
2. Transform each sigma point through nonlinear function
3. Compute Gaussian from the transformed and weighted sigma-points

Unscented Transform
(a.k.a sigma-point transform)

$$UT(\mu, \Sigma) = \begin{cases} x^0 = \mu, & w^0 = \frac{\lambda}{\lambda+n} \\ x^i = \mu + (\sqrt{(\lambda+n)\Sigma})_i, & w^i = \frac{1}{2(\lambda+n)} \quad i = 1, \ldots, n \\ x^{n+i} = \mu - (\sqrt{(\lambda+n)\Sigma})_i, & w^i = \frac{1}{2(\lambda+n)} \quad i = 1, \ldots, n \end{cases}$$

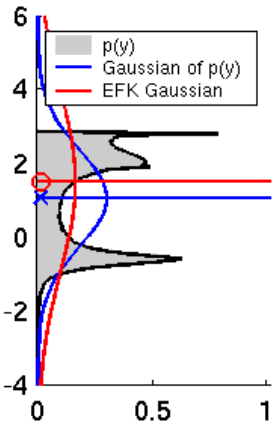


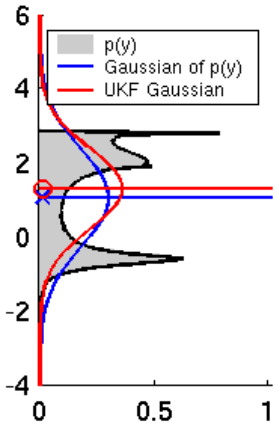
$$UT^{-1}(\{(x^i, w^i)\}_{i=0}^{2}n) = \begin{cases} \mu = \sum_{i=0}^{2n} w^i x^i \\ \Sigma = \sum_{i=0}^{2n} w^i (x^i - \mu)(x^i - \mu)^T \end{cases}$$

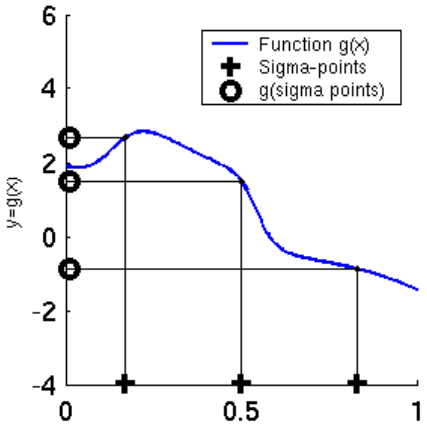
Unscented Transform Proof of Invertibility

Next lecture!

UKF: example

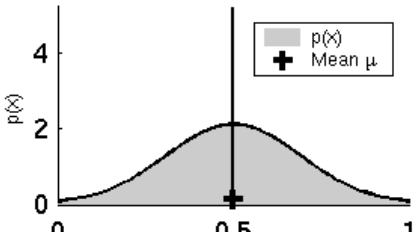




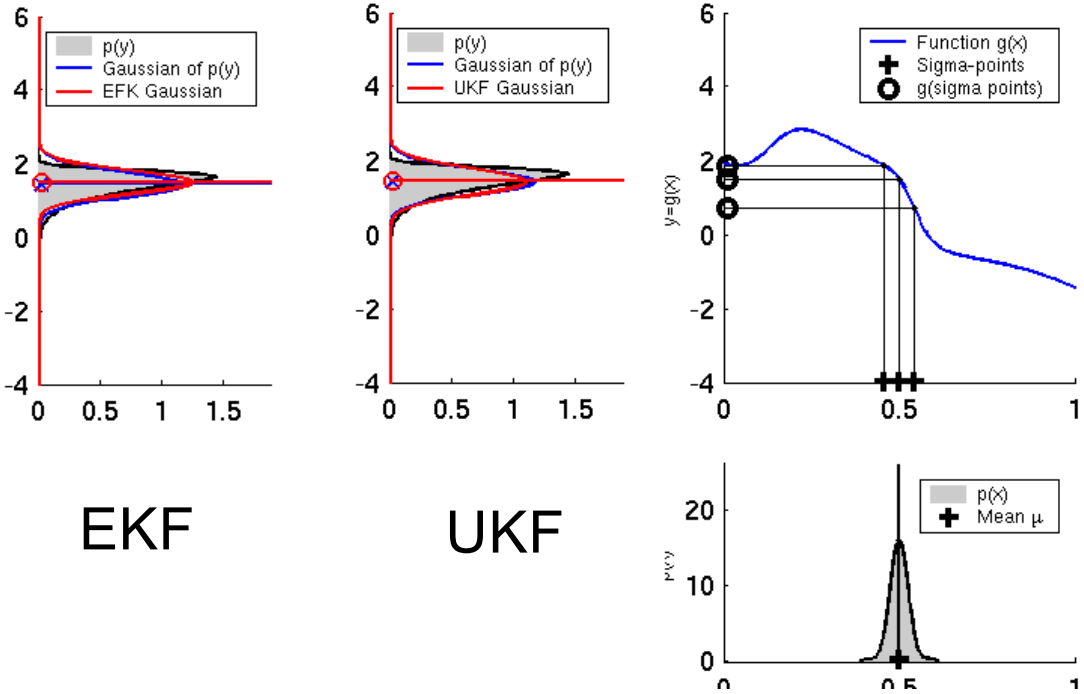


EKF

UKF



UKF: example


p(y)
Gaussian of p(y)
EFK Gaussian
p(y)
Gaussian of p(y)
UKF Gaussian
Function g(x)
Sigma-points
g(sigma points)
y=g(x)
p(x)
Mean μ
EKF
UKF

UKF Algorithm

Given: $\mu_{0|0},\ \Sigma_{0|0},\ f(\cdot,\cdot),\ g(\cdot),\ Q_t,\ R_t$

For: $t = 1, 2, \ldots$

Predict:

$$UT(\mu_{t-1|t-1}, \Sigma_{t-1|t-1}) \rightarrow \{(x^i_{t-1|t-1}, w^i)\}_{i=0}^{2n}$$

Predicted Sig-points $\quad \overline{x}^i_{t|t-1} = f(x^i_{t-1|t-1}, u_{t-1})$

w^i no change

$$\mu_{t|t-1} = \sum_{i=0}^{2n} w^i \overline{x}^i_{t|t-1}$$

Add process noise Cov

$$\Sigma_{t|t-1} = \sum_{i=0}^{2n} w^i (\overline{x}^i_{t|t-1} - \mu_{t|t-1})(\overline{x}^i_{t|t-1} - \mu_{t|t-1})^T + Q_{t-1}$$

Prediction Mean, Cov $\quad (\mu_{t|t-1}, \Sigma_{t|t-1})$

Update:

$$UT(\mu_{t|t-1}, \Sigma_{t|t-1}) \rightarrow \{(x^i_{t|t-1}, w^i)\}_{i=0}^{2n}$$

Predicted Sig-measurements $\quad y^i_{t|t-1} = g(x^i_{t|t-1})$

Predicted mean measurement $\quad \hat{y}_{t|t-1} = \sum_{i=0}^{2n} w^i y^i_{t|t-1}$

Add measure noise Cov

Meas. Cov $\quad \Sigma_y = \sum_{i=0}^{2n} w^i (y^i_{t|t-1} - \hat{y}_{t|t-1})(y^i_{t|t-1} - \hat{y}_{t|t-1})^T + R_t$

Cross Cov $\quad \Sigma_{xy} = \sum_{i=0}^{2n} w^i (x^i_{t|t-1} - \mu_{t|t-1})(y^i_{t|t-1} - \hat{y}_{t|t-1})^T$

Gaussian Est Eqns $\quad \mu_{t|t} = \mu_{t|t-1} + \Sigma_{xy}\Sigma_y^{-1}(y_t - \hat{y}_{t|t-1})$

$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{xy}^T$

Posterior Mean, Cov $\quad (\mu_{t|t}, \Sigma_{t|t})$